Research proposal

Individual proposal template in four parts, built in Sessions 1–7 and ending with a publication plan

Foundations of Clinical Research · Template version 1.1

Proposal title: *[Working title (see 1.1)]*
Author: *[Your name (profession)]*
Department: *[Department]*, *[Institution]*
Version / date: *[Version 1]* · *[DD/MM/YYYY]*

How to use this template

- Replace every [grey italic text in brackets] with your own text. Press Ctrl + H and search for "[" to step through them, and check none remain before you share the document.
- Grey italic guidance and shaded "Example" boxes are there to help you. Delete them as you write: Home → Styles → right-click "Guidance" (or "Example") → Select All → Delete.
- In each proposal builder the instructor fills in one part for the running example, and you then write the same part for your own question at home: Part 1 in Sessions 1 and 2, Part 2 in Sessions 3 and 4, Part 3 in Sessions 5 and 6, Part 4 in Session 7. Everyone submits the proposal in Session 8. Keep it to about 6 to 8 pages.
- The shaded examples are those model answers, the same as on the slides. As on the slides, the running example (fluids in septic shock) changes version: Part 1 uses the pilot trial, Parts 2 and 4 the observational version (a retrospective cohort) and Part 3 the trial with 28-day mortality.
- Choose your own question from your department. Use the examples to see what a good answer looks like, not as text to copy.

Part 1. Ask: the research question (Sessions 1–2)

Running example in Part 1: the pilot trial (version A from Session 1).

1.1 Working title

Build it from your PICO in 10 to 16 words: the population, the intervention or exposure, the comparison and, ideally, the design. No abbreviations and no questions. It is a WORKING title: you refine it in the publication plan (4.5). Written in Session 2.

Example: *Restrictive versus usual fluids in adults with septic shock: a pilot randomised trial*

[Your title]

1.2 Problem statement and rationale

About 300 words in three paragraphs: what is known, what is not known (the gap), then your question and objectives. Cite every fact by number [1]. Never put your own data or results here: a local figure may appear only if it has been published, with its reference. Written in Session 2.

Example, known: *septic shock is a common cause of ICU death. Early fluid resuscitation restores perfusion, but continued fluid often produces a positive fluid balance, which observational studies associate with higher mortality (e.g. Hyun 2023).*

> **Example, not known:** *large trials of restrictive fluids (CLASSIC 2022; CLOVERS 2023) found no mortality difference, and a pilot targeting hidden "non-resuscitative" fluids (Boulet 2024) did not significantly reduce fluid balance at day 5. Whether a restrictive protocol reliably reduces fluid balance is still unclear.*
>
> **Example, question and objectives:** *in adults with septic shock, does a restrictive fluid protocol, compared with usual care, reduce cumulative fluid balance at day 5? Primary objective: to compare day-5 fluid balance between the two groups.*

[What is known, what is not known, your question and objectives]

1.3 PICO

Break your question into its parts. For observational studies, "I" is the exposure (PECO). T (time) is the time point at which the outcome is measured.

	Element	Our study	Example (running example)
P	Population / patients	*[Who?]*	*Adults with septic shock within 24 h of starting vasopressors*
I	Intervention / exposure	*[What is done or what are they exposed to?]*	*Restrictive fluid protocol*
C	Comparison	*[Compared with what?]*	*Usual care*
O	Outcome	*[What will you measure?]*	*Cumulative fluid balance (ml/kg)*
T	Time	*[When?]*	*At day 5*

1.4 Research question

One sentence, ending with a question mark, that contains all PICO(T) elements.

> **Example:** *In adults with septic shock in your ICU, does a restrictive fluid protocol, compared with usual care, reduce cumulative fluid balance at day 5?*

[Our research question]

1.5 FINER check

Is the question worth answering, and can your department answer it? Give each letter a colour and a reason. Be honest: a red letter is a reason to narrow or change the question, not a failure.

	Criterion	Question to ask	Green / Amber / Red	Comment
F	Feasible	Enough patients, time, skills, money and data?	☐ Green ☐ Amber ☐ Red	*[Why?]*
I	Interesting	Do clinicians in your unit want the answer?	☐ Green ☐ Amber ☐ Red	*[Why?]*
N	Novel	Does it confirm, refute or extend what is known (see 1.2)?	☐ Green ☐ Amber ☐ Red	*[Why?]*
E	Ethical	Can it be done without undue risk to patients?	☐ Green ☐ Amber ☐ Red	*[Why?]*
R	Relevant	Would the answer change practice, policy or future research?	☐ Green ☐ Amber ☐ Red	*[Why?]*

> **Example:** *F green/amber (pilot of about 50 patients over 12–18 months) · I green · N amber to green (local setting, different protocol) · E amber (deferred or surrogate consent, Session 7) · R green.*

1.6 Aim, objectives and hypothesis

Aim = the overall purpose (one sentence). Objectives = 2–3 specific, measurable steps (start with a verb: to compare, to estimate, to describe), with the primary one marked. Hypothesis = H0 and H1 for the primary objective, with the same outcome and time point. A descriptive question needs no hypothesis.

Aim: *[To ...]*

Primary objective: *[To compare / estimate ...]*

Secondary objectives: *[1. To ... 2. To ...]*

Null hypothesis (H_0): *[There is no difference in ...]*

Hypothesis (H_1): *[There is a difference in ...]*

> **Example, aim:** *To compare a restrictive fluid protocol with usual care in adults with septic shock in your ICU.*
>
> **Example, objectives:** *(Primary) To compare cumulative fluid balance (ml/kg) at day 5 between groups. To estimate recruitment rate and protocol adherence (feasibility). To describe 28-day mortality and kidney injury needing dialysis in each group (exploratory).*
>
> **Example, hypothesis:** *H0: cumulative fluid balance at day 5 is the same in both groups. H1: it differs.*

1.7 Five key references

The five most relevant papers from your PubMed search in Session 2 (literature review worksheet), in Vancouver style, numbered in the order they are cited.

> **Example:** *Boulet N, Quenot J-P, Serrand C, et al. Impact on fluid balance of an optimized restrictive strategy targeting non-resuscitative fluids in intensive care patients with septic shock: a single-blind, multicenter, randomized, controlled, pilot study. Crit Care. 2024;28:429. doi:10.1186/s13054-024-05155-z*

The other four model references from Session 2: Hyun 2023 (cohort), CLASSIC 2022 and CLOVERS 2023 (large trials), and the Surviving Sepsis Campaign guidelines 2021.

1. *[Reference 1: Authors. Title. Journal. Year;Volume:Pages. doi]*
2. *[Reference 2: Authors. Title. Journal. Year;Volume:Pages. doi]*
3. *[Reference 3: Authors. Title. Journal. Year;Volume:Pages. doi]*
4. *[Reference 4: Authors. Title. Journal. Year;Volume:Pages. doi]*

5. *[Reference 5: Authors. Title. Journal. Year;Volume:Pages. doi]*

Part 2. Design: how the study will be done (Sessions 3–4)

Running example in Part 2: the observational version, a retrospective cohort of fluid balance and 28-day mortality.

2.1 Study design and justification

Tick one design, then explain in 3 to 5 sentences why it is the best feasible design for your question (use the design decision tree from Session 3). Say whether you may write "caused" or only "associated with".

☐ Cross-sectional ☐ Case-control ☐ Prospective cohort ☐ Retrospective cohort ☐ Randomised controlled trial ☐ Other: *[specify]*

> **Example:** *Retrospective cohort. Our question is about the effect of a fluid strategy on death; an RCT is the ideal design but is not feasible for one ICU without funding. Existing charts record daily fluid input and output, severity scores and outcomes, so a cohort can estimate the association between a positive fluid balance and death, and give the effect size for a future trial. We will say "associated with", not "caused".*

[Our design and why]

2.2 Setting

Where (type of unit and hospital), when (study period) and which data source.

> **Example:** *Adult mixed ICU of one teaching hospital; patients admitted over 3 calendar years; data from the electronic chart and the ICU register.*

[e.g. The [number]-bed adult ICU of [Institution], from [month/year] to [month/year]]

2.3 Study population

3 to 5 inclusion criteria (your P, made measurable) and 3 to 5 exclusion criteria, each with a reason. Exclusion criteria remove people who meet the inclusion criteria but should not take part. They are not the inclusion criteria written in the negative.

Inclusion criteria	Exclusion criteria
Age 18 or over	*Chronic dialysis (the fluid balance cannot be interpreted)*
Septic shock (Sepsis-3 definition) on ICU admission	*Transfer from another ICU after more than 24 h (early fluid not recorded here)*
Vasopressor started within 24 h of admission	*Treatment-limitation decision within 24 h*
ICU stay of 72 h or more	*Missing fluid charts on days 1-3*
[Inclusion 1]	*[Exclusion 1]*
[Inclusion 2]	*[Exclusion 2]*
[Inclusion 3]	*[Exclusion 3]*

2.4 Intervention or exposure, and comparator

Describe exactly what the intervention (trial) or exposure (observational study) is, and what it is compared with, so that another unit could do the same.

Intervention / exposure: *[What, how much, how often, who gives it, for how long]*

Comparator: *[e.g. usual care as decided by the treating team]*

> **Example:** *Exposure: positive cumulative fluid balance (ml/kg) at 72 h after ICU admission. Comparator: zero or negative cumulative balance at 72 h.*

2.5 Outcomes

ONE primary outcome (it decides the sample size) and 2 or 3 secondary outcomes. For each: what, how it is measured and when.

Type	Outcome	How measured / definition	When
Primary	*Death from any cause*	*Hospital record (phone call if discharged)*	*By day 28 after ICU admission*
Secondary	*Acute kidney injury*	*KDIGO criteria*	*By day 7*
Secondary	*Days alive and free of mechanical ventilation*	*ICU chart*	*By day 28*
Secondary	*ICU length of stay*	*ICU register*	*At ICU discharge*
Primary	*[Outcome]*	*[Definition]*	*[Time point]*
Secondary	*[Outcome]*	*[Definition]*	*[Time point]*
Secondary	*[Outcome]*	*[Definition]*	*[Time point]*

2.6 Main threats of bias and how they are reduced

Bias and confounding are taught in Session 4, and this section is modelled in Proposal builder 4. Name 2–3 biases that could give a wrong answer in YOUR study and a concrete step against each (randomisation, blinding, matching, adjustment, standard definitions, double checks).

Bias	How it could affect our result	What we will do
Confounding by indication	*Sicker patients receive more fluid*	*Record SOFA score, lactate, vasopressor dose, age and comorbidity at admission and adjust or match*
Selection bias	*Requiring 72 h of ICU stay excludes early deaths*	*State it as a limitation and report how many were excluded and why*
Information bias	*Incomplete fluid charts, missing night-time urine output*	*Pre-specified definitions, two data collectors with a double-entry check on a 10% sample*
[Bias 1]	*[Effect]*	*[Action]*
[Bias 2]	*[Effect]*	*[Action]*

> **Example, claim:** *“associated with” (observational design).*

Part 3. Measure: data and analysis (Sessions 5–6)

Running example in Part 3: the trial of restrictive versus usual fluids, with death by day 28 as the primary outcome.

3.1 Variables table

List every variable you need: the outcome, the exposure or intervention, 3 to 5 confounders and a few descriptive variables, each with how it is measured and its type (continuous, binary, nominal, ordinal, date).

Role	Variable	How measured	Type
Primary outcome	*Death by day 28*	*Hospital records / phone call at day 28*	*Binary*
Secondary outcome	*Cumulative fluid balance at day 5*	*Fluid charts, ml/kg of admission weight*	*Continuous*
Intervention	*Restrictive vs usual fluid strategy*	*Randomised allocation*	*Binary*
Baseline / confounders	*Age, SOFA score, lactate at admission, source of sepsis*	*CRF at ICU admission*	*Continuous, ordinal-like score, continuous, nominal*
[Role]	*[Variable]*	*[How measured]*	*[Type]*
[Role]	*[Variable]*	*[How measured]*	*[Type]*
[Role]	*[Variable]*	*[How measured]*	*[Type]*
[Role]	*[Variable]*	*[How measured]*	*[Type]*

3.2 Data collection tool: mini-CRF (6–10 items)

Write each question exactly as it will appear on the case report form: what, when, unit, format and an option for missing. One item per question, coded tick boxes, dates as DD/MM/YYYY, and a participant ID instead of a name.

#	Question (CRF item)	Answer format and codes
1	*Participant ID*	|__|__|-|__|__|__| *(site, then number)*
2	*Age at ICU admission*	|__|__|__| *years*
3	*Body weight at ICU admission*	|__|__|__| *kg*
4	*Cumulative fluid balance at day 5*	|__|__|__| *ml/kg (+ or −)* ☐ *not recorded*
5	*Vital status at day 28*	☐ *0 Alive* ☐ *1 Dead* ☐ *9 Unknown*
6	*[Item]*	*[Format, unit, codes]*
7	*[Item]*	*[Format, unit, codes]*
8	*[Item]*	*[Format, unit, codes]*

3.3 Data dictionary (key rows)

3 to 5 rows, one per variable in your dataset. Short variable names without spaces, a code for "unknown" (e.g. 9) instead of a blank, and the allowed values, which become range checks.

Variable name	Label (meaning)	Type	Unit / codes	Allowed values
fb_d5	*Cumulative fluid balance, day 5*	*Continuous*	*ml/kg*	*-100 to 300*
src	*Main source of infection*	*Nominal*	*1 lung, 2 urine, 3 abdomen, 4 other, 9 unknown*	*1 to 4, 9*
died28	*Died by day 28*	*Binary*	*0 = alive, 1 = died*	*0, 1*
[name]	*[label]*	*[type]*	*[units]*	*[range]*
[name]	*[label]*	*[type]*	*[units]*	*[range]*

3.4 Data management and quality plan

How will data travel from the patient to the analysis file, and how will you keep them complete and correct? Think: who collects; paper CRF or electronic data capture (e.g. REDCap, KoBoToolbox); training and a CRF pilot; range and consistency checks; double entry of a sample; audit trail; back-ups and version control; where data are stored and for how long; a data-sharing statement.

Step	Our plan
Collection and tool	*[e.g. trained ICU nurses, daily, paper CRF then electronic database]*
Entry and checks	*[e.g. range/consistency checks; 10% double entry; weekly query list]*
Storage, back-up, access	*[e.g. institution server, daily back-up, named users with passwords]*
Data sharing	*[e.g. de-identified data available on reasonable request after publication]*

> **Example:** *EDC with range and logic checks at entry (or paper with double entry of key variables); coded study IDs with the linking log kept separately and locked; audit trail/cleaning log; raw export read-only, dated versions, encrypted 3-2-1 backups; data-sharing statement in the consent form.*

3.5 Sample size statement

Use the sample-size spreadsheet from Session 6. Give the four inputs and where each comes from, then the drop-out allowance. If the number is not feasible, say so and propose a pilot or a multicentre study.

To detect a difference in *[primary outcome]* from *[expected value in comparison group]* to *[expected value in intervention group]*, with 80% power and a 5% significance level (two-sided), we need *[n]* participants per group. Allowing for *[10]*% loss to follow-up, we will recruit *[N]* participants in total.

> **Example:** *Assuming 28-day mortality of 40% with usual care and 30% with the restrictive strategy, 356 patients per group give 80% power at a two-sided alpha of 5%. Allowing 10% loss to follow-up, we will recruit 396 per group (792 in total).*

3.6 Analysis plan in plain language

For each objective: which variable, which summary (mean (SD), median (IQR), n (%)), and which test or model, with the effect measure and its 95% CI (use the test decision table from Session 6). Observational designs: name the confounders chosen in advance (from 2.6), about one model term per 10 events.

Objective	Variable(s)	Summary	Test / method
Describe the patients (Table 1)	*Age, sex, SOFA, lactate*	*Mean (SD) or median (IQR); n (%)*	*None (descriptive)*
Compare 28-day mortality	*died28 by group*	*n (%) per group*	*Chi-square test; risk ratio and risk difference with 95% CIs*
Compare fluid balance	*fb_d5 by group*	*Median (IQR) per group*	*Mann–Whitney U test*
[Objective]	*[Variables]*	*[Summary]*	*[Test]*
[Objective]	*[Variables]*	*[Summary]*	*[Test]*

> **Example, observational version:** *add "adjusted for age, SOFA and lactate using logistic regression".*

3.7 Missing-data plan

Written BEFORE you see the data: which variables are likely to be missing and why, how you will prevent it (Session 5), how you will report it (number missing for each variable, in whom) and the main handling (complete case if little is missing, otherwise multiple imputation).

Prevention: *[e.g. mandatory fields, daily checks while the patient is in the ICU, phone follow-up]*

Reporting: *[e.g. number (%) missing for every variable, in Table 1 and the flow diagram]*

Main analysis: *[e.g. complete-case if < 5% missing; otherwise multiple imputation]*

Sensitivity analysis: *[e.g. best/worst case for missing vital status]*

Part 4. Protect and publish (Session 7)

Running example in Part 4: the observational version again, a retrospective cohort of fluid balance and 28-day mortality in adults with septic shock in our ICU.

4.1 Ethics and consent

Which ethics committee will review the study? What are the risks and benefits for participants? Who gives consent, who asks, and when? In the ICU many patients cannot consent: say whether you will ask a legal representative, use deferred consent, or request a waiver, and why.

Ethics committee: *[Ethics committee name]*

Risks / benefits: *[Main risks, how they are reduced; expected benefits]*

Consent approach: ☐ Written, by the patient ☐ Legal representative ☐ Deferred consent ☐ Waiver requested

Why this approach: *[Justification, referring to [National regulations]]*

> **Example:** *The protocol will be submitted to the institutional research ethics committee before any data collection. Data are collected from routine ICU charts with no change in care. We will request a waiver of individual consent because most patients lack capacity on admission and the study carries no more than minimal risk. If the committee requires consent, it will be sought from the patient when able or from the legally authorised representative.*

4.2 Registration

Trials must be registered in a public registry BEFORE the first participant is enrolled, or most journals will refuse the paper (ICMJE). Registration is recommended for observational studies. Systematic reviews are registered in PROSPERO.

Registry: ☐ ClinicalTrials.gov ☐ A WHO ICTRP primary registry ☐ PROSPERO (reviews) ☐ Other / not required

When / number: *[Planned date (before first participant)] · [Registration ID]*

> **Example:** *Not mandatory for an observational study; recommended (e.g. ClinicalTrials.gov).*

4.3 Confidentiality

How will you protect participants' identity? Think: study ID instead of name, a separate locked key file, password-protected database, who has access, how long data are kept.

> **Example:** *Each patient receives a study code (FLU-001 …). Names, hospital numbers and dates of birth are recorded only on a key file stored on the institution's password-protected drive, accessible to the principal investigator and one data manager. The analysis file contains the study code, age, admission month and study variables only.*

[Our confidentiality measures]

4.4 Feasibility

> **Example:** *The unit admitted about 180 adults with septic shock last year; we expect to enrol 120 eligible patients in 12 months. Two nurses per shift have been trained on the CRF; data are checked weekly. Budget headings: staff time for data entry, printing, open-access fee.*

Timeline (Gantt chart)

Shade the months for each activity (select the cells → Table Design → Shading). Ethics approval usually takes longer than expected.

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Ethics approval												
CRF pilot and staff training												
Recruitment and data collection												
Data cleaning and analysis												
Writing and submission												
[Activity]												
[Activity]												

Budget headings

List the costs, even if the hospital covers them. Only include tests that are not part of routine care.

Budget heading	What it covers	Amount
Personnel	*[e.g. research nurse or data collector time]*	*[0]*
Training	*[e.g. GCP and CRF training]*	*[0]*
Materials and printing	*[e.g. CRFs, consent forms]*	*[0]*
Tests or equipment	*[only if not routine care]*	*[0]*
Data management and statistics	*[e.g. database, statistician time]*	*[0]*
Ethics and registration fees	*[as charged by the committee / registry]*	*[0]*
Dissemination	*[e.g. open-access fee, conference]*	*[0]*
Contingency (5–10%)	*[unexpected costs]*	*[0]*
Total (*[currency]***)**		*[0]*

Resources and recruitment

Take the recruitment figure from your unit's admissions, not from the sample-size calculation.

Question	Our answer
How many eligible patients per month? (source of the number)	*[e.g. 15/month, ICU register]*
How many will agree / be included?	*[e.g. about 70%]*
Who collects the data, and when?	*[Names / roles]*
Are the data routinely recorded?	*[Yes / partly / no]*
What skills or help do we still need?	*[e.g. statistician]*

4.5 Publication plan

Plan the paper now, before any data exist. Choose a journal whose scope and readers fit your question (check it is indexed and not predatory, and its open-access fee), the reporting guideline you will follow from day one, who the authors are, and what your abstract will say.

Target journal and reporting guideline

Target journal: *[First choice]* (back-up: *[second choice]*)

Why this journal: *[Scope and readers; indexed (e.g. PubMed); open access / fee; not predatory]*

Article type: *[e.g. original research, research letter, short report]*

Reporting guideline: ☐ CONSORT (trial) ☐ STROBE (observational) ☐ STARD (diagnostic) ☐ PRISMA (review) ☐ Other

Other channels: *[e.g. department meeting, conference abstract/poster, summary for staff]*

> **Example, target journal:** *a critical-care or critical-care-nursing journal indexed in MEDLINE (checked in the NLM Catalog), APC and licence known; back-up: a general critical-care open-access journal listed in DOAJ.*
>
> **Example, reporting guideline:** *STROBE (cohort), not CONSORT.*
>
> **Example, conference:** *national critical-care congress abstract after the first analysis.*

Authorship and CRediT roles

Authors must meet all four ICMJE criteria: substantial contribution; drafting or critical revision; final approval; accountability. Others go in the Acknowledgements. CRediT roles: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Author (order)	Profession	CRediT roles	Meets ICMJE 4?
1. PI	*[Profession]*	*Conceptualization; Methodology; Writing – original draft*	☐ *Yes*
2. and 3.	*Two nurses*	*Investigation; Data curation*	☐ *Yes*
4.	*Statistician*	*Formal analysis*	☐ *Yes*
[Name]	*[Profession]*	*[Roles]*	☐ Yes
[Name]	*[Profession]*	*[Roles]*	☐ Yes
[Name]	*[Profession]*	*[Roles]*	☐ Yes

> **Example, head of department:** *Supervision only if all four ICMJE criteria are met, otherwise Acknowledgements.*

Working title of the paper

> **Example:** *Cumulative fluid balance and 28-day mortality in adults with septic shock: a retrospective cohort study*

[Refined title from 1.1: population, intervention/exposure, outcome, design]

Structured-abstract skeleton

About 250 words in the final paper. Write the Results row as what you EXPECT to report (which numbers, which effect measure), not invented results.

Heading	Our skeleton	Example (running example)
Background	*[Problem and gap in 1–2 sentences; the aim]*	*Positive fluid balance may be linked to death in septic shock; local data are lacking; we aimed to assess its association with 28-day mortality.*
Methods	*[Design, setting, participants, intervention/exposure, primary outcome, analysis]*	*Retrospective cohort, adults with septic shock, one ICU, 12 months; exposure = cumulative fluid balance at day 3; outcome = 28-day mortality; logistic regression adjusted for severity.*
Expected results	*[Which numbers you will report]*	*"n patients; 28-day mortality x% vs y%; adjusted OR (95% CI)"*
Conclusion	*[What you will be able to say, matched to the design]*	*If positive: "A positive day-3 fluid balance was associated with higher 28-day mortality". If not: "We found no evidence of an association; the CI does not exclude a clinically relevant effect".*

Before you submit in Session 8

- All four parts completed. No [placeholders], grey guidance or shaded examples left.
- The question, objectives, primary outcome, sample size and analysis plan all match each other.
- About 6–8 pages; references in Vancouver style.
- Publication plan completed: target journal, reporting guideline, authors and CRediT roles, working title and abstract skeleton.
- 6 slides prepared from this proposal (see Presentation_Guide).
- Name the file Surname_Firstname_Proposal.docx (e.g. Nguyen_An_Proposal.docx) and email it with your slides to myluong1710@gmail.com before 19:30 on the day of Session 8.